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chelated ring, was less efficient than (**1a**). These findings demonstrate the uniqueness of the functionalized phosphine ligand (**1a**) and suggest that intramolecular co-ordination of the pyridyl group to the nickel atom to form a six-membered chelated ring may be responsible for its effectiveness. The bicyclic 2-pyrone (**3a**) with a fused five-membered ring was similarly produced from (**2a**) and CO₂ by the Ni(cod)₂-(**1a**) catalyst.

The ligand (**1a**) was also effective for the cycloaddition of the terminally monoalkyl-substituted diyne (**2c**) with CO₂, which proceeded regioselectively to give exclusively 5,6,7,8-tetrahydro-1-isopropyl-3*H*-2-benzopyran-3-one (**3c**) without formation of the corresponding 4-isopropyl-substituted regio-isomer. This is synthetically significant and suggests^{6,7} that CO₂ reacts first with the unsubstituted C≡C bond of (**2c**) through the intermediacy of the zero-valent nickel atom.

Coupled with our previous work,⁶ the present study using the parent unsubstituted diynes indicates that Ni(0)-catalysed cycloaddition of diynes and CO₂ provides a general and convenient method of preparing bicyclic 2-pyrones in one step.

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References

- 1 For example, 'Comprehensive Organometallic Chemistry,' vol. 8, eds. G. Wilkinson, F. G. A. Stone, and E. W. Abel, Pergamon, New York, 1982.
- 2 For example, H. B. Kagan, in 'Comprehensive Organometallic Chemistry,' vol. 8, eds. G. Wilkinson, F. G. A. Stone, and E. W. Abel, Pergamon, New York, 1982, p. 463.
- 3 J. C. Jeffrey and T. B. Rauchfuss, *Inorg. Chem.*, 1979, **18**, 2658; M. Bressan, F. Morandini, and P. Rigo, *J. Organomet. Chem.*, 1983, **247**, C8; E. Lindner, H. Rauleder, and P. Wegner, *Z. Naturforsch., Teil B*, 1984, **39**, 1224; S.-E. Bouaoud, P. Braunstein, D. Grandjean, D. Matt, and D. Novel, *Inorg. Chem.*, 1986, **25**, 3765, and references therein.
- 4 The ligand (**1a**) was prepared according to the method used for the preparation of phosphinoacetonitriles, see O. Dahl and F. K. Jensen, *Acta Chem. Scand., Ser. B*, 1975, **29**, 863.
- 5 (a) J. Staunton, in 'Comprehensive Organic Chemistry,' vol. 8, ed. P. G. Sammes, Pergamon, Oxford, 1979, p. 225; (b) G. P. Ellis, in 'Comprehensive Heterocyclic Chemistry,' vol. 3, eds. A. J. Boulton and A. McKillop, Pergamon, Oxford, 1984, p. 645; (c) R. K. Dieter and J. R. Fishpaugh, *J. Org. Chem.*, 1983, **48**, 4439, and references therein; (d) M. E. Jung, J. A. Lowe, M. A. Lyster, M. Node, R. W. Pfluger, and R. W. Brown, *Tetrahedron*, 1984, **40**, 4751.
- 6 T. Tsuda, S. Morikawa, R. Sumiya, and T. Saegusa, *J. Org. Chem.*, 1988, **53**, 3140. See also T. Tsuda, R. Sumiya, and T. Saegusa, *Synth. Commun.*, 1987, **17**, 147.
- 7 H. Hoberg, D. Schaeffer, G. Burkhart, C. Krüger, and M. J. Romao, *J. Organomet. Chem.*, 1984, **266**, 203.